

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 10 2026

Binary and Denary Conversion

*Due Tuesday 8 September, start of the lesson. Show your working where a question asks for it. Every binary number on this sheet is **8 bits** — write all eight digits, including leading zeros. **Total / 17***

1) **Convert** each 8-bit binary number into denary.

[4]

8-bit binary	Denary
00001101	13
00110010	50
10000001	129
11101100	236

2) **Convert** each denary number into 8-bit binary.

[4]

Denary	8-bit binary
6	00000110
40	00101000
100	01100100
173	10101101

3) **Convert** the denary number 221 into 8-bit binary. Show your working.

[2]

221 – 128 = 93 93 – 64 = 29 29 – 16 = 13 13 – 8 = 5 5 – 4 = 1 1 – 1 = 0

128 1 64 1 32 0 16 1 8 1 4 1 2 0 1 1

Answer: **11011101**

MS: 1 mark for the working (repeated division by 2, or place values written with the binary below), 1 mark for 11011101.

4) **Complete** the table by writing in the missing value in each row.

[3]

Denary	8-bit binary
19	00010011
96	01100000
255	11111111

5) **State** the range of denary numbers that can be represented by an 8-bit binary number.

[2]

(a) The **largest** denary number: 255

[1]

(b) The **smallest** denary number: 0

[1]

6) **Describe** how to convert an 8-bit binary number into denary. Use an example in your answer.

[2]

Write the place values 128, 64, 32, 16, 8, 4, 2, 1 above the eight digits, then add together every place value that has a 1 beneath it.

Example: 00110010 gives $32 + 16 + 2 = 50$. *MS: 1 mark for the method (place values, then sum those with a 1), 1 mark for a correct worked example.*